THRIFTKART / SEMESTER III

Project Plan

WBS, network logic, Gantt and resource allocation

Planning basis	Five case-provided work packages + derived dependencies
Calendar constraint	Exam weeks 9 and 10 = zero team capacity
Team	4 students × 3 hours/day on weekdays
Capacity conversion	6 productive hours = 1 day
Schedule target	Front-load work before exam weeks
Milestones	M1–M4, zero-duration checkpoints

1. Work Breakdown Structure

WBS	Work package	Output	Role
1.0	Project management	Scope, requirements, evidence control	PM / BA
1.1	Requirements baseline	BRD, SRS, MoSCoW, traceability	BA
1.2	UML + data model	Use case, class, sequence, activity, state, schema	Design
1.3	Listing & search	Marketplace discovery and listing flow	Development
1.4	Chat	Listing-scoped buyer/seller conversation	Development
1.5	Payments	Payment method/status workflow	Development
1.6	Admin moderation	Approval/rejection and report queue	Admin / Development
1.7	Testing	BVA, equivalence, decision table, integration, evidence	QA
1.8	Closure	Defect metrics, risk closure, final documents	PM / QA

2. Dependency Network

Node	Predecessor(s)	Successor(s)	Logic
A — Listing & search	—	B, D	Foundation for marketplace and moderation branches
B — Chat	A	C	Buyer/seller coordination follows listing discovery
C — Payments	B	E	Payment status belongs to the sale path
D — Admin moderation	A	E	Moderation must be complete before testing the public listing path
E — Testing	C, D	—	System testing begins after the two feeding branches finish

3. Forward Pass, Float & Critical Path

Using the case-provided TE values as activity durations and measuring in productive person-days, the early-start forward pass is: ES(A)=0; EF(A)=6.67; ES(B)=6.67; EF(B)=12.17; ES(C)=12.17; EF(C)=16.84; ES(D)=6.67; EF(D)=10.00; ES(E)=max(16.84,10.00)=16.84; EF(E)=21.17.

Node	Duration	ES	EF	Float	Critical?
A — Listing & search	6.67	0.00	6.67	0.00	Yes
B — Chat	5.50	6.67	12.17	0.00	Yes
C — Payments	4.67	12.17	16.84	0.00	Yes
D — Admin moderation	3.33	6.67	10.00	6.84	No
E — Testing	4.33	16.84	21.17	0.00	Yes

Critical path

A → B → C → E = 21.17 productive person-days. Under the team capacity of 2 productive person-days per calendar day, that is approximately 10.59 working days, or 2.12 calendar weeks. Admin moderation has 6.84 person-days of float in this network.

4. Three-Week Gantt View

ID	Item	Week 1	Week 2	Week 3	Note
M1	Requirements & scope baseline	◆	—	—	—
A	Listing & search	■■■■	—	—	Critical
D	Admin moderation	■■	—	—	Float branch
M2	Design baseline	◆	—	—	—
B	Chat	■■■	■■	—	Critical
C	Payments	—	■■■	—	Critical
M3	Feature complete	—	◆	—	—
E	Testing	—	■■	—	Critical
M4	Assessment release	—	◆	—	—

The Gantt is a derived presentation of the forward-pass logic rather than an additional set of invented task durations. The key scheduling rule is to finish the quantitative work before exam weeks 9 and 10, when capacity is zero.

5. Resource Allocation

Role	Primary responsibility	Concurrent focus
Student 1 — BA / PM	BRD, SRS, MoSCoW, scope, mentor evidence	Coordinates requirements and closure
Student 2 — UI / Frontend	Marketplace UI, listing screens, responsive prototype	Supports listing/search and presentation
Student 3 — Backend / Data	API boundary, validation, schema, transaction state	Supports listing, payments, moderation
Student 4 — QA / UML	UML package, test design, defect metrics	Supports moderation rules and evidence

6. Milestone Gate Criteria

Milestone	Gate	Exit evidence
M1	Scope frozen	Approved BRD/SRS baseline + MoSCoW + traceability structure
M2	Design frozen	UML diagrams + relational schema agree with SRS
M3	Feature complete	Core listing/search, moderation and sale workflow ready for system test
M4	Assessment release	Test evidence, metrics, risks and final documents packaged

7. Schedule Assumptions

- The case-provided TE values are expected-effort estimates, not guaranteed durations.
- Calendar conversion assumes the full team capacity can be allocated to the active work package; real student coordination can reduce usable capacity.
- Weeks 9 and 10 are treated as zero-capacity exam weeks, so no planned development work is placed there.
- The project should use the critical path as the primary monitoring signal and use the moderation branch float to absorb small slips.